

CHALMERS



Finite Element Method - Structures, TME245

Computer assignment 1b: Elastoplastic analysis of a chairlift hook

Instructions

Martin Fagerström

Department of Mechanical Engineering
CHALMERS UNIVERSITY OF TECHNOLOGY
TME245 FEM - Structures 2025/2026

Preface

Within this computer assignment, a 2D plane strain plasticity problem is considered. The expected learning outcomes of this assignment are to be able to implement a six-noded isoparametric triangular element with numerical integration, and to be able to implement a solver for nonlinear finite element problems exemplified with an elastoplastic analysis.

All derivations need to be well structured and easy to read (written by hand or using a word-processor). Also, submit your MATLAB (or Python) code as part of the hand-in (Abaqus files are not to be submitted).

The project work is to be carried out in groups with preferably, and **at most, two students in each group**. The students should contribute equally to solving the tasks. **Please write names, group number and course code clearly on the front page of the report and submit it with attachments on CANVAS.**

Since the projects prepare you for the examination, it is essential that each group solves the tasks separately and practice writing code as well as doing any necessary derivations. To clarify,

It is OK to:

- Collaborate and discuss around derivations asked for in different subtasks. HOWEVER, each group should in the end do the derivations themselves and hand-in their own written solutions.
- Discuss around implementation strategies in MATLAB for different subtasks. HOWEVER, limit the discussions to pen and paper otherwise you will know that you are on the border of what is not OK.

It is NOT OK to:

- Directly copy other groups' derivations and submit them as your own.
- Share any amount of MATLAB-code or files from an FE software (e.g. Abaqus) between groups.
- Copy and submit code that is partly written by another group. This is true for code written by other students this year as well as from previous years.

Any suspected cheating (including things like copying MATLAB or Python code or parts of someone else report) will be directly reported to the Disciplinary Committee.

Göteborg in January 2022
Martin Fagerström

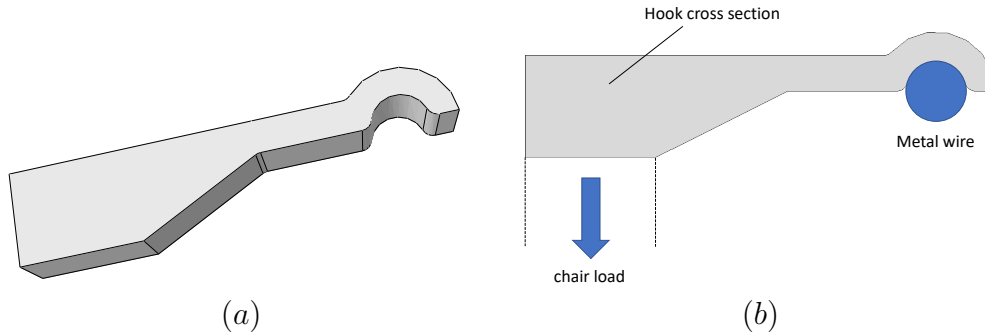


Figure 1: Illustration of the chair hook section to be analysed in Computer Assignment 1b. Please note that the thickness of the hook section is 60 mm across its entire cross section. (a): 3D geometry; (b): Schematic of loading case, including the hook section (in light grey), the metal wire (in blue) and an indication of the remaining chair structure that provides the load to the hook part. Please note that only the grey section is to be analysed.

Problem definition

Chairlifts as shown on the front page are commonly used to transport skiers up the mountain. Each chair often has 2-6 seats, and is made from metal frame firmly connected to a metal wire that runs in a loop up and down the mountain.

In this task, we will analyse the connection part between the metal wire and the chair, namely the hook section as illustrated in grey in Figure 1. Please note that this hook is clamped to the wire (which is not to be included in the model), such that no relative movement between hook and wire can occur.

In Task 1 you will do a 3D elastoplastic analysis of the hook using Abaqus. In Task 2, you will analyse the same hook using your own code (MATLAB or Python), but under a simplifying 2D plane strain assumption. More details related to each task can be found below.

Hook material behaviour

The hook section is made from forged aluminium, with elastic properties $E = 80 \text{ GPa}$ and $\nu = 0.2$, and is expected to deform plastically with isotropic hardening with the following plastic properties: yield stress $\sigma_{\text{yield}} = 170 \text{ MPa}$ and isotropic hardening modulus $H = 225 \text{ MPa}$.

Task 1: Elastoplastic analysis in 3D using Abaqus

In this task, a 3D section as provided via the Abaqus CAE file `hook3D.cae` is to be analysed. Please note that the file only contains the definition of the hook section geometry, with all dimensions in mm.

Please also note that you are expected to produce a load displacement plot (see subtasks (c) and (d)) that shows the response of the hook section under increasing load on the chair, starting from a completely unloaded state of 0 N. It will be up to you to define appropriate boundary conditions, with the aim to be as representative of the load case as illustrated in Figure 1(b).

Please complete the following subtasks.

- (a) Define appropriate boundary conditions for the problem to be analysed. Make your own sketch of the hook section and clearly define and motivate your choice of boundary conditions.

Please note that the increasing load is best applied as an increasing prescribed displacement (as opposed to directly imposing an increasing load). Prescribed displacements will most likely lead to easier convergence in the Newton iterations. Therefore, the subtask will be to identify a suitable part of the boundary surface where you can apply an increasing prescribed displacement, and to identify another part of the boundary surface where you constrain the hook section from moving (in a suitable way).

- (b) From the geometrical model in `hook3D.cae`, define the complete finite element model in terms of the following steps:
- **Define the aluminium elastoplastic behaviour** (see hint at the end of this task)
 - **Create a section and assign that to the part**
 - **Create an instance of the part in the assembly module**
 - **Create a static load case.** Automatic time stepping over a quasi-time period from 0 to 1 should suffice to obtain convergence. However, if that is not the case, or if you want to follow the deformation and stress evolution in more steps, you can adjust the initial time step and maximum increment size under the 'Incrementation' tab.
 - **Define boundary conditions according to what you specified in subtask (a)**
 - **Discretise the domain in a suitable way** using the element type of your choice. Make sure to motivate the type of discretisation (size and element type) in the report. Later, you all also analyse what is required to obtain converged results.

- **Create and run a job.**

Please make sure to briefly explain how you did each of these steps in the report.

- (c) **Run the model up to a prescribed vertical displacement of a magnitude of 5 mm**, and produce the following results (to be included in the report):

- **A contour plot of the von Mises stress distribution in the section.** It is sufficient to display stresses on the outside surface. However, should you also want to make sections and look inside the hook section volume, you can do that via `Tools -> View Cut -> Manager....`
- **A contour plot that illustrates which parts of the domain that experience plastic deformation.** This is most easily done by post-processing the equivalent plastic strain `PEEQ`, setting the contour limits (`Options -> Contour -> Limits`) to a very small value (of your choice).
- **A graph illustrating the force-displacement relation at the boundary where you have the varying prescribed vertical displacement.** The force should be computed as the sum of all vertical nodal reaction forces along that part of the boundary (see pdf on Canvas for instructions).

- (d) **Rerun the analysis with finer and finer discretisation (smaller and smaller elements) until your force-displacement response has converged.** State what element size that is required for the element type you have chosen. And explain how you have determined that the results have converged.

Some Abaqus hints:

1. Check links to useful video clips and instruction pdfs for Abaqus available on Canvas.
2. To obtain the elastoplastic behaviour as defined by yield stress $\sigma_{\text{yield}} = 170$ MPa and isotropic hardening modulus $H = 225$ MPa, please note that you can define the isotropic hardening in Abaqus terms of a yield stress at zero plastic strain of 170 MPa, and a yield stress of 215 MPa at 0.2 plastic strain, cf. the figure below.

Edit Material

Name: aluminium

Description:

Material Behaviors

Elastic

Plastic

General

Mechanical

Thermal

Electrical/Magnetic

Other

Plastic

☐ Use scale stress value

Hardening:

Isotropic

☐ Use strain-rate-dependent data

☐ Use temperature-dependent data

Number of field variables:

0

Suboptions

Data

	Yield Stress	Plastic Strain
1	170	0
2	215	0.2

OK

Cancel

Task 2: Elastic and Elastoplastic analysis in 2D using MATLAB (or Python if you prefer)

For the second task, we will now make a simplified 2D analysis of a cross-section of the hook under the assumption of plane strain (remember that in this case, the out-of-plane thickness is 60 mm).

To do so, you will first adapt your linear elasticity code from CE1 to the geometry and load case for the hook section. Then, in a second step you will replace the linear triangular elements with 6-noded quadratic triangular elements. In the third step you will re-write the code into a nonlinear finite element programme, however still considering a linear elastic material response. Finally, you will replace the linear elastic material behaviour with the proper elastoplastic behaviour as indicate above.

In more detail, please solve the following subtasks:

- a) Use your elasticity code from Computer Exercise 1 to **solve for the displacement of the hook section under the assumption that the aluminium behaves linear elastic**. Consider the magnitude of the prescribed vertical displacement to be 1 mm.

Geometry and element topology for a predefined discretised geometry can be found on Canvas in the file `hook2D_coarse_linear.mat`. Please note that all coordinates are expressed in millimetres. By using the topology provided, define your choice of additional matrices/vectors for keeping track of what degrees of freedom that the boundary conditions are subjected to. Check that mesh looks correct by using `eldraw2(Ex,Ey,[1 2 0])`.

Please note that a few node sets have been defined to provide nodes along certain parts of the boundary. The names of these sets should be fairly self-explanatory.

Compute and report the total vertical reaction force along the moving boundary displaced a magnitude of 1 mm in the vertical direction (obtained by summation of vertical nodal reaction forces along that boundary).

- b) Modify your elasticity code in the way that you **replace the constant strain triangles with quadratic isoparametric 6-noded triangular elements** (see Lecture notes for theory - refer also to quadrilateral isoparametric elements for useful code snippets).

For this task, geometry and element topology can be found on Canvas (see the file `hook2D_coarse.mat`). Please note that again all coordinates are expressed in millimetres. To check your geometry, please plot the undeformed mesh using `eldraw2_ext.m` (also provided on Canvas).

For this task, you need to write your own element routine, including isoparametric mapping and numerical integration (with three integration points). Make sure your element routine has both the element internal force vector and an (elastic) element stiffness matrix as output.

Solve the same problem and comment on the possible difference in the resulting reaction force. In case there is a difference in results, even though the number of elements is identical, please explain why. Also, provide a plot of the deformed mesh with 6-noded triangular elements. For this, you can use `eldisp2_ext.m` available on Canvas.

To check your code, you can use the following test case. For an element with thickness $h = 10$ mm and the nodal coordinates:

$$\underline{x}^e = \begin{bmatrix} 1.0 \\ 1.0 \\ 0.0 \\ 1.0 \\ 0.5 \\ 0.5 \end{bmatrix} \text{ mm and } \underline{y}^e = \begin{bmatrix} 0.0 \\ 0.5 \\ 0.5 \\ 0.25 \\ 0.5 \\ 0.25 \end{bmatrix} \text{ mm}$$

that has the following values of the element degrees of freedom

$$\underline{a}^e = \begin{bmatrix} -14.014 \\ 0.0000 \\ -14.014 \\ 28.028 \\ 0.0000 \\ 28.028 \\ -14.014 \\ 14.014 \\ -7.0070 \\ 28.028 \\ -7.0070 \\ 14.014 \end{bmatrix} \cdot 10^{-5} \text{ mm},$$

the element internal force vector should be

$$\underline{f}^{e,\text{int}} = \begin{bmatrix} 1.5860 \cdot 10^{-13} \\ -77.856 \\ -6.8669 \cdot 10^{-13} \\ 77.856 \\ -2.9447 \cdot 10^{-13} \\ -7.9494 \cdot 10^{-16} \\ -1.6032 \cdot 10^{-14} \\ -8.8818 \cdot 10^{-14} \\ -8.9864 \cdot 10^{-13} \\ 311.42 \\ 1.7372 \cdot 10^{-12} \\ -311.42 \end{bmatrix} \text{ N}$$

Please note that the very small numerical values may differ to some extent, but if your implementation is correct you should be very close.

- c) As the next task, **re-write your elasticity code as a nonlinear finite element programme**. As such, your code must include a step-wise increase of the magnitude of the prescribed boundary displacements as well as an iterative Newton method to solve for the updated free displacement degree of freedoms for each load step.

For this task, it is recommended that you start from the nonlinear truss structure code that you have already been given, and then make necessary changes to accommodate for the 2D problem with 6-noded triangular elements.

Verify that, since we are still considering linear elastic behaviour, the Newton iterations still converge in a maximum of two iterations per load step. Nothing needs to be reported in this step, but it is essential to be able to solve the subsequent subtasks.

- d) Now, we introduce elastoplastic behaviour of the aluminium as described above.

Modify your element routine for the 6-noded isoparametric triangles to account for that the material is now elastoplastic. Please note that the elastoplastic stress response can be computed using the function `mises`, and that the tangent material behaviour $\partial \underline{\sigma} / \partial \underline{\epsilon}$ can be computed by the function `dmises`.

Here, it is important to note that the elastic trial stress at Newton iteration k , that is asked for as input to `mises`, is given by:

$${}^{(k)}\underline{\sigma}^{\text{trial}} = {}^n\underline{\sigma} + \underline{D} {}^{(k)}\Delta \underline{\epsilon}$$

with ${}^n\sigma$ being the stress from the previous time step and ${}^{(k)}\Delta\epsilon$ the estimated increment in strain from the previous to the current time step ${}^{(k)}\Delta\epsilon = {}^{(k)}\epsilon - {}^n\epsilon$. Please also note that σ and ϵ in this context also contain the out-of-plane normal components (which however are not part of computing the internal force vector)

Solve the problem with a suitable number of load steps, dividing the loading up to a maximum prescribed vertical displacements of magnitude 5 mm. **Report what you find as a suitable and robust way to increase the amount of boundary displacement.** Please note that, if you use equally spaced load steps, you may need to divide the analysis into as many as 600 load steps to guarantee convergence.

- e) For the given discretisation, **compute and plot how the total vertical force develops as a function of the increasing vertically prescribed displacement** (you need to explicitly save these during the analysis).